

Name: _____

Date: _____

Solve each problem. Show your work in the box.

Write your final answer on the answer line (or in the blanks shown).

Use the strategies you learned in Lesson 5.08.04.

1 Volume of a Complex Shape

A solid is made of two cuboids joined together: Cuboid A: $6\text{ cm} \times 4\text{ cm} \times 3\text{ cm}$ Cuboid B: $6\text{ cm} \times 2\text{ cm} \times 5\text{ cm}$

What is the total volume?

Total volume: _____

2 Volume of a Complex Shape

An L-shaped solid has outer dimensions $10\text{ in.} \times 8\text{ in.} \times 4\text{ in.}$ A $5\text{ in.} \times 3\text{ in.} \times 4\text{ in.}$ cuboid is removed from one corner.

What is the remaining volume? Show your work.

Volume: _____

3 Decompose

A T-shaped solid can be split into a top cuboid ($8\text{ cm} \times 2\text{ cm} \times 3\text{ cm}$) and a bottom cuboid ($3\text{ cm} \times 2\text{ cm} \times 5\text{ cm}$).

Find the total volume.

Total volume: _____

4 Applied

A step is built from two rectangular blocks. The bottom block is $50\text{ cm} \times 30\text{ cm} \times 10\text{ cm}$. The top block is $50\text{ cm} \times 15\text{ cm} \times 10\text{ cm}$.

What is the total volume of the step?

Volume: _____

5 Reasoning

When finding the volume of a complex solid, explain why you must be careful not to count any region twice. Use an example.